

Sean Luka Girgis

Python / AWS Developer | Financial Services, SDLC, Testing & Production Support

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Professional Summary

Senior Python, AWS, and financial-services technology professional with 20+ years of enterprise experience across Python, SQL, AWS data/platform workflows, Oracle/MySQL concepts, SDLC delivery, requirements analysis, design documentation, testing, production support, troubleshooting, REST/SOAP integration concepts, and AI-assisted development. Strong background translating business and technical needs into reliable software/data workflows, documenting designs and support procedures, validating outputs, resolving production issues, and collaborating with cross-functional teams in regulated financial-services environments. Best aligned to Python/AWS consulting roles requiring hands-on development, SDLC discipline, testing, code review concepts, database development, production reliability, documentation, and client/stakeholder communication; modern web technologies, Java, JavaScript, and broad full-stack application ownership are positioned carefully as adjacent or earlier-career/working-knowledge areas rather than overstated current production ownership.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Built and supported Python/Pandas/PySpark and SQL workflows ingesting telemetry from 6,000+ infrastructure endpoints into curated datasets for reporting, analytics, forecasting, validation, and operational decision support.
- Developed AWS and warehouse-backed platform components using S3, AWS Glue, Athena, Amazon Redshift, Oracle, SQL, and Python/PySpark patterns to support scalable ingestion, transformation, querying, and analytics.
- Supported end-to-end SDLC-style delivery by clarifying requirements, designing data flows, implementing code changes, validating outputs, documenting behavior, and coordinating fixes with application, infrastructure, and business teams.
- Troubleshoot production issues by tracing source feeds, Python processing, SQL transformations, metric behavior, failed outputs, latency symptoms, and downstream reporting discrepancies.
- Implemented testing and validation checks for row counts, nulls, duplicates, unmatched joins, data freshness, abnormal deltas, and output comparisons against trusted prior runs.
- Created runbooks, source mappings, conceptual data flows, metric definitions, validation checkpoints, support procedures, and technical documentation to enrich team knowledge bases and improve handoff.
- Used AI-assisted development tools including ChatGPT/Copilot-style workflows and GenAI/RAG-oriented project work to accelerate documentation, code review thinking, test generation, and data workflow development.
- Communicated technical and operational issues clearly to technical and non-technical stakeholders in a regulated financial-services environment.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Built analytics and reporting outputs from Dynatrace AppMon and Gomez Synthetic Monitoring data for Brand.com and critical hospitality systems.
- Analyzed application and monitoring data to identify production-like performance issues, reliability risks, data quality issues, and optimization opportunities.
- Created dashboards and reporting views that converted raw monitoring data into actionable performance, reliability, and stakeholder insights.
- Collaborated with application and infrastructure teams to validate findings, document issues, and support practical system improvements.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Supported a large financial-services monitoring and reporting environment with 50+ Enterprise Managers and 4,000–6,000 instrumented agents.
- Provided production troubleshooting guidance to application, middleware, infrastructure, and operations teams using CA APM/Introscope telemetry, dashboards, alerts, and transaction visibility.
- Built Perl and KornShell automation for extracting, validating, transforming, and distributing APM telemetry, replacing manual reporting workflows with repeatable data-processing and support workflows.
- Created technical documentation, support procedures, onboarding material, dashboard standards, alert thresholds, and troubleshooting notes used by multiple teams.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Developed and executed performance test plans for enterprise telecom applications, including baseline comparisons, regression review, and release-readiness analysis.
- Analyzed J2EE application behavior under load to identify throughput limits, SQL/JDBC bottlenecks, CPU, memory, thread, and garbage-collection issues.
- Configured JMX Monitoring, Wily Introscope, and thread-dump automation scripts to improve runtime visibility during testing and production-readiness reviews.
- Documented test results, expected behavior, defects, risks, and remediation recommendations for engineering and release teams.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led migration of a high-throughput shopping engine from 200+ MySQL nodes to a 6-node Oracle RAC cluster while preserving transaction performance and data integrity.
- Built C++/OCCI/OCI validation and regression testing utilities to verify correctness across millions of migrated transaction records.
- Optimized SQL and Oracle access patterns for high-volume workloads while reducing physical hardware footprint by 95%.
- Supported schema conversion, data validation, phased cutover, troubleshooting, documentation, and production-readiness activities for a mission-critical enterprise migration.

Education

Post-Graduate Diploma in Computer Engineering Technology / Computer Science

Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering

Zagazig University, Egypt

Skills

Flagship Projects

Serverless Lakehouse Platform (AWS)

Designed an AWS data platform pattern using S3, Glue, Athena, Redshift, PySpark, and Parquet for raw ingestion, curated transformations, analytics datasets, query optimization, and validation checkpoints.

Technologies: AWS S3, AWS Glue, Athena, Redshift, PySpark, Parquet, SQL

ServiceCall AI RAG Demo

Built staged Python proof-of-concepts with document loading, chunking, TF-IDF retrieval, hybrid retrieval, Pydantic schemas, FastAPI service layers, Docker-first smoke tests, and deterministic API endpoints.

Technologies: Python, FastAPI, Docker, Pydantic, pytest, RAG, TF-IDF

HorizonScale — Capacity Forecasting and Data Platform

Built Python/Pandas/PySpark workflows to process telemetry, validate outputs, prepare forecasting datasets, and produce trusted capacity and risk reporting for financial-services infrastructure.

Technologies: Python, Pandas, PySpark, SQL, Prophet, scikit-learn, Data Validation

Enterprise Application Monitoring and Support

Supported production monitoring, troubleshooting, dashboarding, and documentation across CA APM, Dynatrace, AppDynamics, JMX, and financial-services enterprise applications.

Technologies: CA APM, Dynatrace, AppDynamics, JMX, Documentation, Troubleshooting